

Polar Curves & Curvature

Module 1 Study Guide – Understanding, Not Just Memorizing

Based on your SJCIT lecture notes (22MATM11 / Mathematics-I)

The Roadmap

Everything in this module builds on *one* idea: relating an angle in a polar curve (ϕ , the angle between the radius vector and the tangent) to the calculus of the curve ($dr/d\theta$). Once you have that one relation, four other results fall out of it almost mechanically:

1. $\tan \phi = r d\theta/dr$ – the foundational relation (derived first).
2. **Angle between two curves** – just find ϕ for each curve, then subtract.
3. **Pedal equation** – rewrite the same ϕ relation in terms of p (perpendicular distance from pole to tangent) instead of θ .
4. **Radius of curvature** – a completely different quantity (how sharply the curve bends), derived in four different coordinate systems.
5. **Arc length** – a supporting formula used inside curvature derivations.

Master step 1 conceptually, and steps 2–3 become almost automatic. Step 4 is a separate skill (differentiation-heavy) but follows a fixed template every time.

1 Polar Coordinates – The Setup

Instead of locating a point by (x, y) , polar coordinates locate it by *how far* it is from a fixed origin (r) and *what angle* the line to it makes with a fixed direction (θ). A curve is then $r = f(\theta)$ instead of $y = f(x)$.

Key labeled quantities you'll see in every diagram:

- $O(0, 0)$ – the **pole** (origin of the polar system).
- $P(r, \theta)$ – a point on the curve; $r = OP$ is the **radius vector**.
- θ – angle between the radius vector OP and the fixed initial line OX .
- T – the tangent line to the curve at P .
- ϕ – angle between the radius vector OP and the tangent T . **This is the star of the whole module.**
- ψ – angle the tangent T makes with the initial line OX (used only as a stepping stone to get ϕ).

2 Deriving $\tan \phi = r \frac{d\theta}{dr}$

The tangent's direction (ψ , measured from the fixed x -axis) can be computed two ways: (a) directly, using $\tan \psi = dy/dx$ like in ordinary Cartesian calculus, since polar coordinates are secretly built from $x = r \cos \theta$, $y = r \sin \theta$; and (b) geometrically, since $\psi = \phi + \theta$ (the tangent's angle from the axis is the sum of the angle from the radius vector, plus the radius vector's own angle from the axis). Setting these two expressions for $\tan \psi$ equal and untangling the algebra gives you ϕ in terms of r and θ .

Logical flow of the derivation (this is exactly your notes' Page 2–4):

1. State the geometric fact: $\psi = \phi + \theta$, so $\tan \psi = \tan(\phi + \theta) = \frac{\tan \phi + \tan \theta}{1 - \tan \phi \tan \theta}$.
2. Separately, since $x = r \cos \theta$ and $y = r \sin \theta$, differentiate both with respect to θ and form $\tan \psi = \frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta}$.
3. Grind through the algebra (product rule on $x = r \cos \theta$, $y = r \sin \theta$) to get $\tan \psi$ purely in terms of $\tan \theta$ and $r \frac{d\theta}{dr}$.
4. Compare this with step 1's expression term-by-term (both are fractions of the form $\frac{\tan \theta + \square}{1 - \tan \theta \cdot \square}$) – matching the $\square$ in

both gives $\tan \phi = r \frac{d\theta}{dr}$.

Common mistake: Students often try to memorize the algebra step-by-step instead of understanding *why* we compare two expressions for $\tan \psi$. If you understand that both routes compute the *same* angle ψ , the "why" of comparing them becomes obvious, and you can reconstruct the algebra even if you forget a step.

Remember: The equivalent form $\cot \phi = \frac{1}{r} \frac{dr}{d\theta}$ is usually more convenient for problems since most given curves are $r = f(\theta)$, so you differentiate r directly and divide by r – no need to invert.

3 Angle Between Two Polar Curves

If two curves cross at a point P , each curve has its own tangent there, and each tangent makes its own angle (ϕ_1 for curve 1, ϕ_2 for curve 2) with the *same* radius vector OP . The angle between the two curves at P is simply the angle between their two tangents, which is $|\phi_2 - \phi_1|$.

Procedure (applies to every "find the angle between curves" problem):

1. Find $\cot \phi_1$ for curve 1 using $\cot \phi_1 = \frac{1}{r} \frac{dr}{d\theta}$.
2. Find $\cot \phi_2$ for curve 2 the same way.
3. Solve each for ϕ_1, ϕ_2 (usually lands as a clean multiple of θ , like $\phi_1 = \theta/2$).
4. Compute $|\phi_2 - \phi_1|$. If both curves are given with a shared θ (i.e. they intersect at a general point), the θ terms often *cancel out completely*, leaving a constant like $\pi/2$ – this is exactly what "show the curves cut orthogonally" is testing.
5. If instead you're asked for the angle at one specific point, first find the point of intersection (solve the two curve equations together for θ), then substitute that θ into your ϕ_1, ϕ_2 expressions.

Remember: Orthogonal means $|\phi_2 - \phi_1| = \pi/2$. In practice, this is equivalent to $\cot \phi_1 \cdot \cot \phi_2 = -1$ (or $\tan \phi_1 \cdot \tan \phi_2 = -1$) – if you don't want to solve for ϕ_1, ϕ_2 individually, you can just check this product shortcut.

Common mistake: When curves are given as $r = a(1 - \cos \theta)$ and $r = b(1 + \cos \theta)$ (different constants a, b), notice the constants a, b actually *cancel out* during the $\cot \phi = \frac{1}{r} \frac{dr}{d\theta}$ step, because $\frac{1}{r} \frac{dr}{d\theta}$ only depends on the *shape* (the θ -dependence), not the size. This is why the orthogonality result $\pi/2$ holds for *any* a, b – a good sanity check on your algebra.

Problem Types You'll See (from your notes)

- Cardioid pairs: $r = a(1 \pm \cos \theta)$, $r = a(1 \pm \sin \theta)$ – always simplify using half-angle identities ($1 + \cos \theta = 2 \cos^2(\theta/2)$, etc.) to get clean $\cot \phi = \cot(\theta/2)$ -type results.
- Conic pairs: $r = \frac{a}{1 + \cos \theta}$, $r = \frac{b}{1 - \cos \theta}$.
- Power curves: $r^n = a^n \cos n\theta$, $r^n = b^n \sin n\theta$ – these always give a constant angle $\pi/2$ between them, independent of n .
- Transcendental: $r = a \log \theta$ vs $r = a / \log \theta$ – these require solving $\log \theta = 1$ (i.e. $\theta = e$) to find the intersection point first, since the curves only meet at a specific θ , not for all θ .

4 Pedal Equation

The "pedal equation" is just a way of describing a curve using p (the perpendicular distance from the pole to the tangent line) and r only – with θ eliminated entirely. Why do this? Because p shows up naturally in physics applications (e.g. central force problems, orbital mechanics) where you care about the perpendicular distance from a force center to the direction of motion.

Derivation logic (your notes, page 5):

1. Drop a perpendicular from the pole O to the tangent line T , meeting it at M ; let $OM = p$.
2. In the right triangle OPM : $\sin \phi = \frac{OM}{OP} = \frac{p}{r}$, so $p = r \sin \phi$. This is the simplest form of the pedal relation.

3. To eliminate ϕ and get a relation purely in r, p (or r, θ), take the reciprocal and square: $\frac{1}{p^2} = \frac{1}{r^2 \sin^2 \phi} = \frac{1}{r^2} \csc^2 \phi = \frac{1}{r^2}(1 + \cot^2 \phi)$.
4. Substitute the Section 3 result $\cot \phi = \frac{1}{r} \frac{dr}{d\theta}$ into this, giving the working formula

$$\frac{1}{p^2} = \frac{1}{r^2} + \frac{1}{r^4} \left(\frac{dr}{d\theta} \right)^2$$

Remember: Every pedal-equation problem follows the exact same three moves: (1) differentiate the given curve to get $dr/d\theta$, (2) plug into the boxed formula above to get $1/p^2$ in terms of r, θ , (3) use the *original* curve equation to eliminate θ so only r, p remain.

Common mistake: Step 3 (eliminating θ) is where most marks are lost – it's easy to get a correct expression in r, θ, p and stop there. The question always wants θ completely gone. Look back at the original curve equation for a substitution that removes it (e.g. if $r = a(1 + \cos \theta)$, notice $1 + \cos \theta = r/a$ and substitute that in).

5 Radius of Curvature – Four Forms, One Idea

Curvature measures how sharply a curve bends – formally, the rate at which the tangent's direction (ψ) changes as you move a small distance ds along the curve: curvature = $d\psi/ds$. The **radius of curvature** ρ is just the reciprocal: $\rho = ds/d\psi$. A gently curving road has large ρ ; a tight hairpin bend has small ρ . Every formula below is the same $\rho = ds/d\psi$ idea, just computed using whatever coordinates the curve is given in.

4a. Cartesian Form: $\rho = \frac{(1 + y_1^2)^{3/2}}{y_2}$

Since $\tan \psi = dy/dx = y_1$, we get $\psi = \tan^{-1}(y_1)$. Differentiating, $\frac{d\psi}{dx} = \frac{y_2}{1 + y_1^2}$. Also $\frac{ds}{dx} = \sqrt{1 + y_1^2}$ (standard arc-length relation). Then

$$\rho = \frac{ds}{d\psi} = \frac{ds/dx}{d\psi/dx} = \frac{\sqrt{1 + y_1^2}}{y_2/(1 + y_1^2)} = \frac{(1 + y_1^2)^{3/2}}{y_2}$$

Use when: y is given directly as a function of x . This is the version you'll use most, since many curves in problems (parabola $y^2 = 4ax$, curves like $x^3 + y^3 = 3axy$, $a^2y = x^3 - a^3$) are easiest left in Cartesian form.

4b. Polar Form: $\rho = \frac{(r^2 + r_1^2)^{3/2}}{r^2 + 2r_1^2 - r r_2}$

Same idea, but now $\psi = \phi + \theta$ (geometric decomposition again), so $\frac{d\psi}{ds} = \frac{d\theta}{ds} \left[\frac{d\phi}{d\theta} + 1 \right]$. Using $\tan \phi = r/r_1$ (from Section 3) and differentiating that relation with respect to θ gives $d\phi/d\theta$ in terms of r, r_1, r_2 . Combine with $ds/d\theta = \sqrt{r^2 + r_1^2}$ (polar arc length) and invert to get ρ .

Use when: the curve is given as $r = f(\theta)$ and converting to Cartesian would be messy (e.g. $r^n = a^n \cos n\theta$, cardioids, spirals).

4c. Parametric Form: $\rho = \frac{(\dot{x}^2 + \dot{y}^2)^{3/2}}{\dot{x}\ddot{y} - \dot{y}\ddot{x}}$

Use when: both x and y are given as functions of a third parameter t (the cycloid $x = a(\theta + \sin \theta)$, $y = a(1 - \cos \theta)$ is the classic example in your notes). This is really the Cartesian formula rewritten using the parametric chain rule, since computing $y_1 = \dot{y}/\dot{x}$ directly and then y_2 from it algebraically simplifies to this cleaner form.

4d. Pedal Form: $\rho = r \frac{dr}{dp}$

Use when: you already have (or can quickly find) the pedal equation from Section 4. This is often the *fastest* route for curves whose pedal equation is already a familiar simple relation like $p^2 = ar$ or $2ap^2 = r^3$ – you skip Cartesian/polar differentiation entirely and just differentiate the pedal relation itself with respect to p .

Remember: All four formulas answer the exact same question (how sharply does the curve bend at this point?) – which one you use is purely a matter of which form the curve was originally given in. Never convert coordinate systems just to force a "preferred" formula; use whichever form matches the given curve.

6 Arc Length – The Supporting Formula

Arc length (s) measures distance *along* the curve rather than straight-line distance. It appears inside every curvature derivation as the ds term. You rarely compute arc length as a final answer in this module – it's a tool used mid-derivation.

$$\begin{array}{ll} \text{Cartesian: } \frac{ds}{dx} = \sqrt{1 + y_1^2} & \text{Parametric: } \frac{ds}{dt} = \sqrt{\dot{x}^2 + \dot{y}^2} \\ \text{Polar: } \frac{ds}{d\theta} = \sqrt{r^2 + r_1^2} & (\text{equivalently}) \frac{ds}{dr} = \sqrt{1 + r^2 \left(\frac{d\theta}{dr}\right)^2} \end{array}$$

How to Study This Module

1. **Don't memorize the final boxed formulas in isolation.** Instead, be able to reproduce the 3–4 line logical chain that gets you there (geometric fact $\rightarrow$ algebraic manipulation $\rightarrow$ result). VTU "prove that" questions are graded on this chain, not just the final answer.
2. **Drill the differentiation, not the concept.** Once you understand $\cot \phi = \frac{1}{r} \frac{dr}{d\theta}$ conceptually, the actual exam difficulty is almost entirely in correctly differentiating curves like $r = a(1 + \cos \theta)$ or $r^n \cos n\theta = a^n$ without algebra slips – especially converting $1 + \cos \theta$ or $1 - \cos \theta$ into half-angle form cleanly.
3. **Recognize the four curvature formulas as "the same idea, different coordinates"** rather than four unrelated things to memorize – this halves your memorization load.
4. **Practice the "eliminate θ " step specifically** for pedal equations – it's a distinct skill from the differentiation itself and is where most partial-credit is lost.

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